Installing the Latest Ansible

Python virtual environment method

Where Ansible gets installed

Ansible is installed on the **control node** only: the machine you run playbooks from. The Keysets are the **managed (target) nodes**. Nothing needs to be installed on them beyond a compatible Python and SSH access (unless the playbooks use raw or network modules).

- **Control node:** Python 3.11+ for the latest ansible-core (steps below).
- **Targets (Keysets):** the newest ansible-core also requires a minimum Python version on targets. Check it with `python3 --version` on a Keyset. If it is older than the supported minimum, pin an older Ansible release (see Notes).

1. Check prerequisites (control node)

```
python3 --version
```

The latest ansible-core needs Python 3.11+ on the control machine. If the system Python is older, install a newer Python alongside it. Do not replace the system Python.

2a. Machine with internet access

```
python3 -m venv ~/ansible-venv
source ~/ansible-venv/bin/activate
pip install --upgrade pip
pip install ansible
```

2b. Isolated network (no internet)

On a connected machine with the **same OS, CPU architecture, and Python version** as the control node:

```
python3 -m venv tmp-venv && source tmp-venv/bin/activate
pip download ansible -d ansible-wheels
```

Copy the `ansible-wheels` folder to the control node, then:

```
python3 -m venv ~/ansible-venv
source ~/ansible-venv/bin/activate
pip install --no-index --find-links=ansible-wheels ansible
```

3. Verify

```
ansible --version
```

4. Using it later

Activate the venv in each new shell before running playbooks:

```
source ~/ansible-venv/bin/activate
ansible-playbook <playbook>.yml
```

5. Collections (if the playbooks need them)

```
ansible-galaxy collection install <namespace.collection>
```

Offline: run the following on a connected machine, copy the tarballs over, then install each file:

```
ansible-galaxy collection download <namespace.collection>
ansible-galaxy collection install <file>.tar.gz
```

Notes

- Target hosts also need a compatible Python (unless using raw or network modules).
- If the Python on the Keysets is too old for the newest Ansible, pin an older release, e.g. `pip install "ansible<12"`. Check the version-to-Python support table in the Ansible docs to pick the right pin.
- Do not use `sudo pip`. Keep everything inside the venv.